

CLASSROOM CONTACT PROGRAMME

(ACADEMIC SESSION 2019-2020)

PRE-MEDICAL : ENTHUSIAST COURSE [PHASE-II]

TEST # 07

DATE : 18-08-2019

Test Type : Minor

Test Pattern : NEET

INSTRUCTIONS

Do not open this Test Booklet until you are asked to do so

1. Immediately fill in the form number on this page of the Test Booklet with Blue/Black Ball Point Pen. Use of pencil is strictly prohibited.
2. The candidates should not write their Form Number anywhere else (except in the specified space) on the Test Booklet/Answer Sheet.
3. The test is of **3 hours** duration.
4. The Test Booklet consists of **180** questions. The maximum marks are **720**.
5. There are **three** parts in the question paper A,B,C consisting of **Physics, Chemistry** and **Biology** having **45 questions** each in Physics & Chemistry and **90 questions** in Biology part of equal weightage. Each question is allotted 4 (four) marks for **correct** response.
6. **One Fourth** mark will be deducted for indicated incorrect response of each question. **No deduction** from the total score will be made if no response is indicated for an item in the Answer Sheet.
7. Use **Blue/Black Ball Point Pen only** for writing particulars/markings responses on **Side-1** the Answer Sheet. **Use of pencil is strictly prohibited.**
8. No candidate is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone any electronic device etc, except the Identity Card inside the examination hall/room.
9. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
10. On completion of the test, the candidate must hand over the Answer Sheet to the invigilator on duty in the Room/Hall. **However, the candidate are allowed to take away this Test Booklet with them.**
11. **Do not fold or make any stray marks on the Answer Sheet.**

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MINOR TEST # 07

Topics

DATE : 18 - 08 - 2019

PRE-MEDICAL ENTHUSIAST COURSE : PHASE - II – PATTERN : [NEET]

PHYSICS : Geometrical Optics, Magnetic Effect of Current

CHEMISTRY : GOC & Hydrocarbons

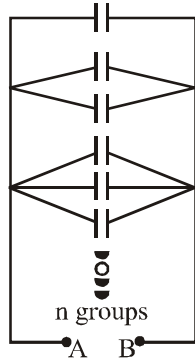
BIOLOGY : DNA to Structure of t-RNA, Reproductive Health (Complete)

SPACE FOR ROUGH WORK

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

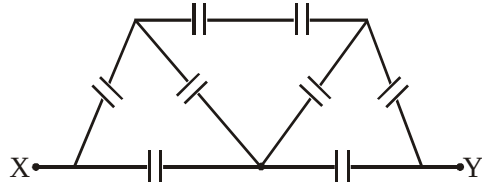
BEWARE OF NEGATIVE MARKING

1. A number of capacitors, each of equal capacitance C , are arranged as shown in Fig. The equivalent capacitance between A and B is:-



- (1) n^2C (2) $(2n + 1)C$
(3) $\frac{(n-1)n}{2}C$ (4) $\frac{(n+1)n}{2}C$

2. Eight capacitors, each of capacity C are connected as shown in fig. The equivalent capacitance between X and Y is :



- (1) $40 C/5$ (2) $16 C/5$
(3) $5 C/6$ (4) $6 C/5$

3. A parallel plate capacitor has plates of area A and separation d and is charged to a potential difference V . The charging battery is then disconnected and the plates are pulled apart until their separation is $2d$. What is the work required to separated the plates?

- (1) $2\varepsilon_0 AV^2/d$ (2) $\varepsilon_0 AV^2/d$
(3) $3\varepsilon_0 AV^2/2d$ (4) $\varepsilon_0 AV^2/2d$

4. For section AB of a circuit shown in Fig, $C_1 = 1\mu F$, $C_2 = 2\mu F$, $E = 10V$, and the potential difference $V_A - V_B = -10V$. Charge on capacitor C_1 is:-

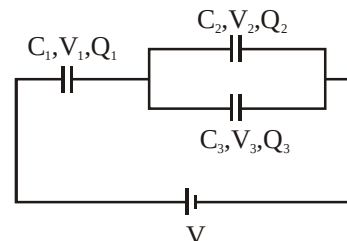


- (1) $0\mu C$ (2) $20/3\mu C$
(3) $40/3\mu C$ (4) none of these

5. Ten capacitors are joined in parallel and charged with a battery up to a potential V . They are then disconnected from the battery and joined in series. Then, the potential of this combination will be:-

- (1) $1V$ (2) $10V$
(3) $5V$ (4) $2V$

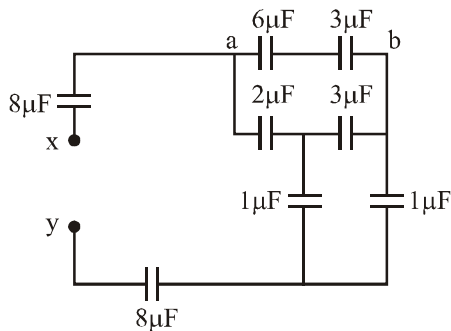
6. In Fig, three capacitors C_1 , C_2 and C_3 are joined to a battery. With symbols having their usual meaning, the correct conditions will be:-



- (1) $Q_1 = Q_2 = Q_3$ and $V_1 = V_2 = V_3 + V$
(2) $Q_1 = Q_2 + Q_3$ and $V = V_1 + V_2 + V_3$
(3) $Q_1 = Q_2 + Q_3$ and $V = V_1 + V_2$
(4) $Q_2 = Q_3$ and $V_2 = V_3$

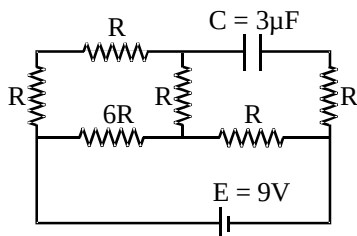
SPACE FOR ROUGH WORK

7. In the circuit shown, the effective capacitance between points X and Y is:-



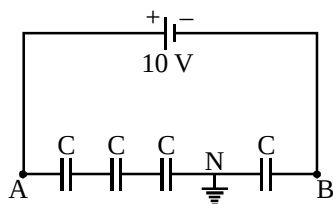
- (1) $3.33 \mu\text{F}$ (2) $1 \mu\text{F}$
(3) $0.44 \mu\text{F}$ (4) none of these

8. In the circuit, the each resistance is of 3Ω . The cell is ideal, potential difference across $3\mu\text{F}$ is :-



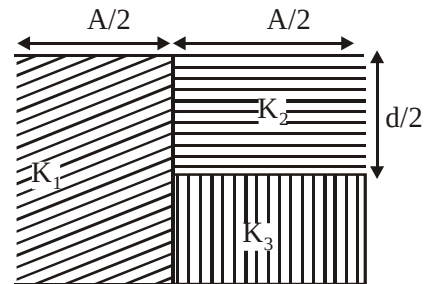
- (1) 4 V (2) 6 V
(3) 8 V (4) 3 V

9. Four identical capacitors are connected in series with a 10 V battery as shown in the figure. The point N is earthed. The potential of points A and B are :-



- (1) 10 V, 0V (2) 7.5 V, -2.5V
(3) 5V, -5V (4) 7.5 V, 2.5 V

10. A parallel plate capacitor of area A, plate separation d and capacitance C is filled with three dielectrics as shown in Fig. The equivalent capacitance is proportional to :-



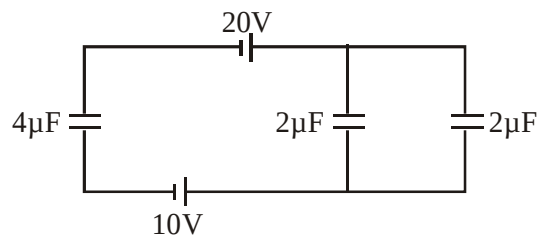
(1) $K = \frac{K_1}{2} + \frac{K_2 K_3}{K_2 + K_3}$

(2) $\frac{1}{K} = \frac{2}{K_1} + \frac{1}{K_2} + \frac{1}{K_3}$

(3) $K = \frac{K_1 K_2}{K_1 + K_2} + 2K_3$

(4) $K = \frac{K_2 K_3}{K_2 + K_3} + 2K_1$

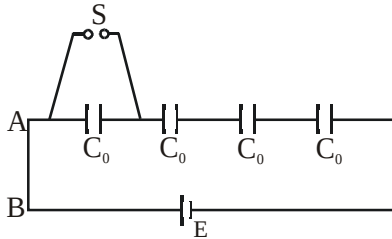
11. In the circuit shown in Fig., charge stored in $4\mu\text{F}$ capacitor is



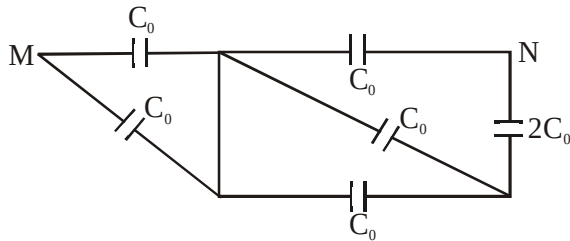
- (1) $20 \mu\text{C}$ (2) $10 \mu\text{C}$
(3) $40 \mu\text{C}$ (4) $120 \mu\text{C}$

SPACE FOR ROUGH WORK

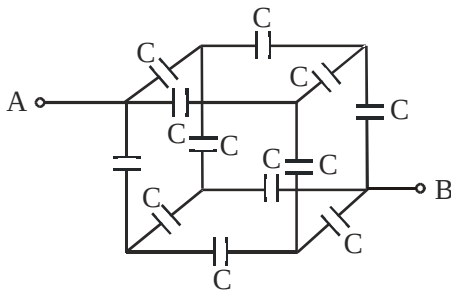
12. In the given figure, find the charge flowing through section AB when switch S is closed:



- (1) $C_0 E/12$ (2) $C_0 E/4$
(3) $C_0 E/3$ (4) none of these
13. The equivalent capacitance between points M and N is:

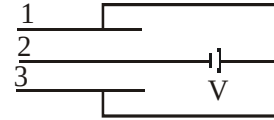


- (1) $\frac{10}{11}C_0$ (2) $2C_0$
(3) C_0 (4) none of these
14. The equivalent capacitance between A and B is: (All are identical capacitor)



- (1) $5C/7$ (2) $7C/5$
(3) $5C/6$ (4) $6C/5$

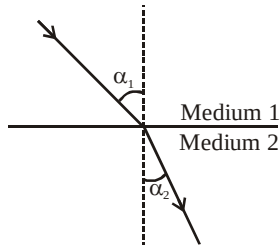
15. Three plates 1, 2 and 3 of area A each and separation between two consecutive plates d, are connected as shown in figure. The energy stored, when the plates are fully charged, is:



- (1) $\frac{0.5\epsilon_0 AV^2}{d}$
(2) $\frac{\epsilon_0 AV^2}{d}$
(3) $\frac{2\epsilon_0 AV^2}{d}$
(4) $\frac{\epsilon_0 AV}{d}$
16. A capacitor is charged by using a battery, which is then disconnected. A dielectric slab is then slid between the plates which results in :
- (1) reduction of charge on the plates and increase of potential difference across the plates
(2) increase in the potential difference across the plates, reduction in stored energy but no change in the charge on the plates
(3) decrease 'in the potential difference across the plates, reduction in stored energy, but no change in the charge on the plates
(4) none of the above

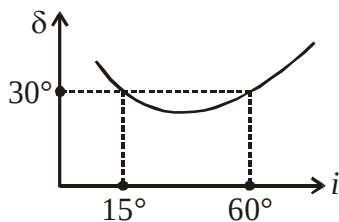
SPACE FOR ROUGH WORK

17. A beam of light propagates through a medium 1 and falls onto another medium 2, at an angle α_1 as shown in Fig. After that, it propagates in medium 2 at an angle α_2 as shown. The light's wavelength in medium 1 is λ_1 . What is the wavelength of light in medium 2 ?



- (1) $\frac{\sin \alpha_1}{\sin \alpha_2}$ (2) $\frac{\sin \alpha_2}{\sin \alpha_1} \lambda_1$
(3) $\frac{\cos \alpha_1}{\cos \alpha_2} \lambda_1$ (4) $\frac{\cos \alpha_2}{\cos \alpha_1} \lambda_1$

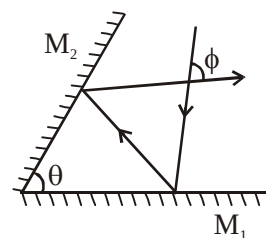
18. Figure shows the graph of angle of deviation δ versus angle of incidence i for a light ray striking a prism. The prism angle is :-



- (1) 30° (2) 45° (3) 60° (4) 75°
19. How will the image formed by a convex lens be affected, if the central portion of the lens is wrapped in black paper, as shown in Fig. ?



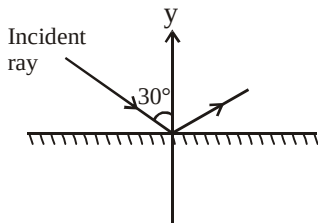
- (1) No image will be formed
(2) Full image will be formed but it is less bright
(3) Full image will be formed but without the central portion
(4) Two images will be formed, one due to each exposed half
20. Calculate the dispersive power for crown glass when $\mu_V = 1.60$ & $\mu_R = 1.50$:-
(1) 0.3 (2) 0.18
(3) 0.4 (4) None
21. The reflecting surfaces of two mirrors M_1 and M_2 are at an angle θ as shown in the figure. A ray of light is incident on M_1 . Total deviation suffered by incident ray is.



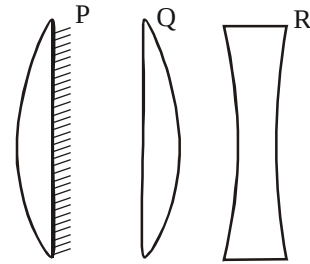
- (1) $180 - 2\theta$ (2) $180 - \theta$
(3) $90 - \theta$ (4) $360 - 2\theta$

SPACE FOR ROUGH WORK

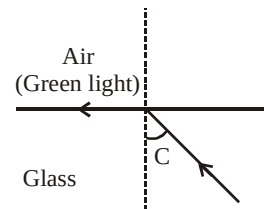
22. By what angle mirror should be rotated to obtain reflected ray on +y-axis :



- (1) 360° CW (2) 15° ACW
(3) 350° CW (4) 16° ACW
23. When an object is kept at a distance of 30 cm from a concave mirror, the real image is formed at a distance of 10 cm. If the object is moved with a speed of 9 m/s along principal axis, the speed with which image moves is :
- (1) 10 m/s (2) 1 m/s
(3) 9 m/s (4) 0.9 m/s
24. Mark the correct options :
- (1) If the reflected rays are converging, we have a real object
(2) If the final reflected rays are converging, we have a real image
(3) The image of a virtual object is called a virtual image
(4) If the image is virtual, the corresponding object is called a virtual object
25. Figure shows three lenses of equal radii of curvature of the curved surfaces and equal refractive index. The ratio of focal lengths of P, Q and R is :-



- (1) 1 : 1 : 1
(2) 1 : 1 : -1
(3) -1 : 2 : 1
(4) -1 : 2 : -1
26. A beam of white light is incident on glass-air interface from glass to air such that green light just suffers total internal reflection. The colors of the light which will come out to air are :-

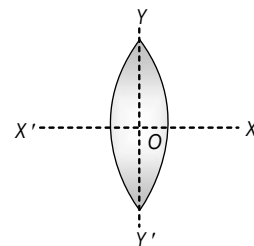


- (1) Violet, Indigo, Blue
(2) All colors except green
(3) Yellow, Orange, Red
(4) White light
27. For Astronomical telescope, focal length of objective & eyepiece are 10 cm & 2 cm respectively. Find magnification of compound of telescope when image is formed at infinity :-
- (1) 25 (2) 5
(3) 15 (4) 20

SPACE FOR ROUGH WORK

28. A point object is placed at the center of a glass sphere of radius 6 cm and refractive index 1.5. The distance of virtual image from the surface is :-
 (1) 6 cm (2) 4 cm
 (3) 12 cm (4) 9 cm
29. A vessel of depth $2d$ cm is half filled with a liquid of refractive index μ_1 and the upper half with a liquid of refractive index μ_2 . The apparent depth of the vessel seen perpendicularly is
 (1) $d \left(\frac{\mu_1 \mu_2}{\mu_1 + \mu_2} \right)$ (2) $d \left(\frac{1}{\mu_1} + \frac{1}{\mu_2} \right)$
 (3) $2d \left(\frac{1}{\mu_1} + \frac{1}{\mu_2} \right)$ (4) $2d \left(\frac{1}{\mu_1 \mu_2} \right)$
30. All of the following statements are correct except
 (1) The magnification produced by a convex mirror is always less than one
 (2) A virtual, erect, same-sized image can be obtained using a plane mirror
 (3) A virtual, erect, magnified image can be formed using a concave mirror
 (4) A real, inverted, same-sized image can be formed using a convex mirror
31. If an object is placed 10cm in front of a concave mirror of focal length 20cm, the image will be
 (1) Diminished, upright, virtual
 (2) Enlarged, upright, virtual
 (3) Diminished, inverted, real
 (4) Enlarged, upright, real

32. A diver in a swimming pool wants to signal his distress to a person lying on the edge of the pool by flashing his water proof flash light
 (1) He must direct the beam vertically upwards
 (2) He has to direct the beam horizontally
 (3) He has to direct the beam at an angle to the vertical which is slightly less than the critical angle of incidence for total internal reflection
 (4) He has to direct the beam at an angle to the vertical which is slightly more than the critical angle of incidence for the total internal reflection
33. An equiconvex lens is cut into two halves along (i) XOX' and (ii) YOY' as shown in the figure. Let f, f', f'' be the focal lengths of the complete lens, of each half in case (i), and of each half in case (ii), respectively

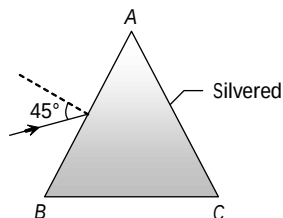


Choose the correct statement from the following

- (1) $f' = 2f, f'' = f$
 (2) $f' = f, f'' = f$
 (3) $f' = 2f, f'' = 2f$
 (4) $f' = f, f'' = 2f$

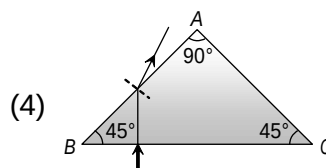
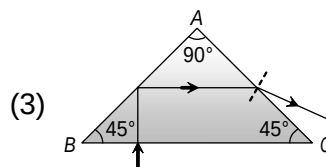
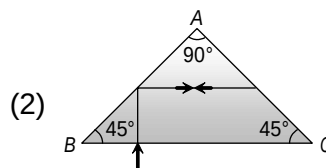
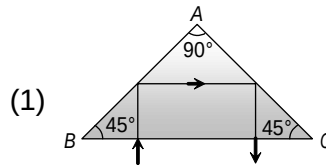
SPACE FOR ROUGH WORK

34. The power of an achromatic convergent lens of two lenses is $+2D$. The power of convex lens is $+5D$. The ratio of dispersive power of convex and concave lens will be
 (1) $5 : 3$ (2) $3 : 5$
 (3) $2 : 5$ (4) $5 : 2$
35. A medium is said to be dispersive, if
 (1) Light of different wavelengths propagate at different speeds
 (2) Light of different wavelengths propagate at same speed but has different frequencies
 (3) Light is gradually bent rather than sharply refracted at an interface between the medium and air
 (4) Light is never totally internally reflected
36. A prism ABC of angle 30° has its face AC silvered. A ray of light incident at an angle of 45° at the face AB retraces its path after refraction at face AB and reflection at face AC . The refractive index of the material of the prism is



- (1) 1.5 (2) $\frac{3}{\sqrt{2}}$
 (3) $\sqrt{2}$ (4) $\frac{4}{3}$

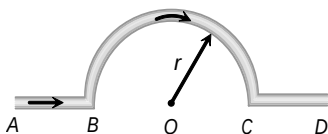
37. The refractive index of a material of a prism of angles $45^\circ - 45^\circ - 90^\circ$ is 1.5. The path of the ray of light incident normally on the hypotenuse side is shown in



38. A man can see the objects upto a distance of one metre from his eyes. For correcting his eye sight so that he can see an object at infinity, he requires a lens whose power is
 (1) $+0.5 D$
 (2) $+1.0 D$
 (3) $+2.0 D$
 (4) $-1.0 D$

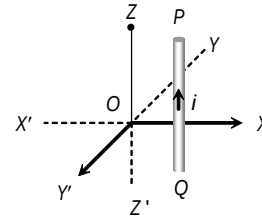
SPACE FOR ROUGH WORK

39. A person is suffering from myopic defect. He is able to see clear objects placed at 15 cm. What type and of what focal length of lens he should use to see clearly the object placed 60 cm away
- Concave lens of 20 cm focal length
 - Convex lens of 20 cm focal length
 - Concave lens of 12 cm focal length
 - Convex lens of 12 cm focal length
40. In order to increase the magnifying power of a compound microscope
- The focal lengths of the objective and the eye piece should be small
 - Objective should have small focal length and the eye piece large
 - Both should have large focal lengths
 - The objective should have large focal length and eye piece should have small
41. The aperture of a telescope is made large, in order to
- To increase the intensity of image
 - To decrease the intensity of image
 - To have both greater magnification and intensity of image
 - To have lesser resolution
42. In the figure, shown the magnetic induction at the centre of the arc due to the current in portion AB will be



- $\frac{\mu_0 i}{r}$
- $\frac{\mu_0 i}{2r}$
- $\frac{\mu_0 i}{4r}$
- zero

43. A vertical wire kept in Z-X plane carries a current from Q to P (see figure). The magnetic field due to current will have the direction at the origin O along



- OX
 - OX'
 - OY
 - OY'
44. The magnetic induction at any point due to a long straight wire carrying a current is
- Proportional to the distance from the wire
 - Inversely proportional to the distance from wire
 - Inversely proportional to the square of the distance from the wire
 - Does not depend on distance
45. The magnetic field $d\vec{B}$ due to a small current element $d\vec{l}$ at a distance $\vec{r}$ and element carrying current i is,

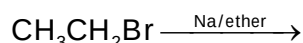
OR

Vector form of Biot-savart's law is

- $d\vec{B} = \frac{\mu_0}{4\pi} i \left(\frac{d\vec{l} \times \vec{r}}{r} \right)$
- $d\vec{B} = \frac{\mu_0}{4\pi} i^2 \left(\frac{d\vec{l} \times \vec{r}}{r} \right)$
- $d\vec{B} = \frac{\mu_0}{4\pi} i^2 \left(\frac{d\vec{l} \times \vec{r}}{r^2} \right)$
- $d\vec{B} = \frac{\mu_0}{4\pi} i \left(\frac{d\vec{l} \times \vec{r}}{r^3} \right)$

SPACE FOR ROUGH WORK

46. Intermediate of the Wurtz reaction is



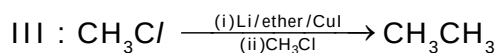
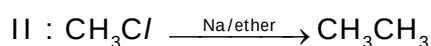
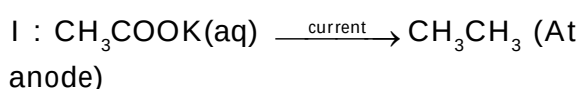
- (1) $\text{CH}_3\dot{\text{C}}\text{H}_2$
- (2) $\dot{\text{B}}\text{r}$
- (3) $\text{CH}_3\text{CH}_2\dot{\text{C}}\text{HCH}_3$
- (4) $\text{CH}_3\text{CH}_2\text{CH}_2\dot{\text{C}}\text{H}_2$

47. $(\text{CH}_3)_3\text{COH} + \text{CH}_3\text{MgBr} \rightarrow \text{Hydrocarbon}$

(A). Then (A) is

- (1) $(\text{CH}_3)_3\text{CH}_3$
- (2) $(\text{CH}_3)_3\text{CH}$
- (3) CH_4
- (4) None of these

48. Consider the following reactions for the formation of alkanes



These reactions are

- | I | II | III |
|-----------|--------|-------------|
| (1) Kolbe | Wurtz | Fittig |
| (2) Kolbe | Fittig | Wurtz |
| (3) Kolbe | Fittig | Corey-House |
| (4) Kolbe | Wurtz | Corey-House |

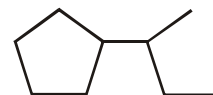
49. R_2CuLi is the intermediate formed in the Corey-House synthesis. It is called

- (1) Corey-House reagent
- (2) Gilman reagent
- (3) Frankland reagent
- (4) Schmidt reagent

50. $\text{C}_5\text{H}_{11}\text{Cl}$ by Wurtz reaction forms 2,2,5,5-tetramethylhexane as the main product. Hence, $\text{C}_5\text{H}_{11}\text{Cl}$ is

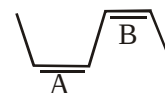
- (1) 2, 2-dimethyl-1-chloropropane
- (2) 2-methyl-1-chlorobutane
- (3) Both (a) and (b)
- (4) None of the above true

51. What is the total number of σ -bonds present in the molecule shown?



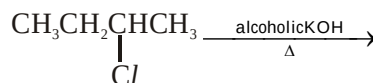
- (1) 18
- (2) 26
- (3) 27
- (4) 30

52. What are the configurations (E or Z) of the indicated positions A and B in the following?



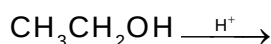
- (1) E, E
- (2) Z, Z
- (3) E, Z
- (4) Z, E

53. Major product of the following reaction is



- (1) 1-butene
- (2) 2-butene
- (3) 2-methyl-1-propene
- (4) None of these

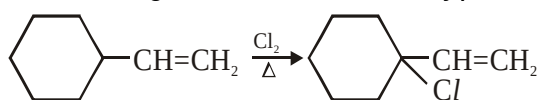
54. What is the possible product in the following reaction ?



- (1) $\text{CH}_2=\text{CH}_2$
- (2) $\text{CH}_3\text{CH}=\text{CH}_2$
- (3) Both of (1) and (2)
- (4) None of these

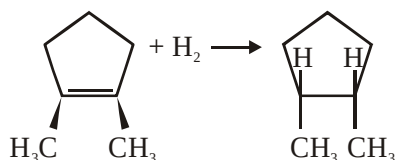
SPACE FOR ROUGH WORK

55. Following reaction is of the type



- (1) S_N^1
(2) S_N^2
(3) E_1
(4) free-radical substitution

56. Consider the following addition reaction



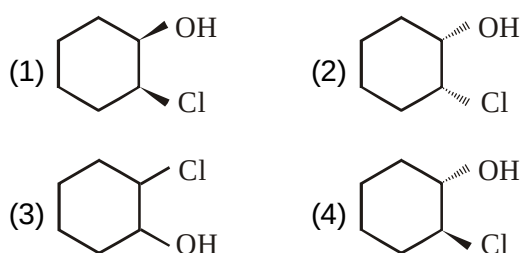
Select the correct statement about this reaction.

- (1) This is a case of syn-addition and compound formed is racemic
(2) This is a case of trans-addition and compound formed is meso
(3) This is a case of syn-addition and compound formed is meso
(4) All of the above are correct statements

57. Meso-compound is formed when Br_2 adds to

- (1) cis-2-butene (2) trans-2-butene
(3) 1-butene (4) any alkene

58. Chlorohydrin formed when $\text{Cl}_2/\text{H}_2\text{O}$ adds to cyclohexene has configuration



59. Consider the following addition reaction.
 $\text{CH}_3\text{CH=CH}_2 + \text{HBr} \rightarrow \text{CH}_3\text{CH}_2\text{CH}_2\text{Br}$
This

- (1) takes place by free-radical mechanism
(2) takes place in presence of peroxide
(3) follows anti-Markownikoff's rule
(4) follows all these statements

60. General name for $:\text{CH}_2$ is

- (1) carbene (2) carbocation
(3) carbanion (4) carbonium ion

61. By oxidation ozonolysis $\text{A}(\text{C}_4\text{H}_8)$ changes to CH_3COOH . Thus, A is

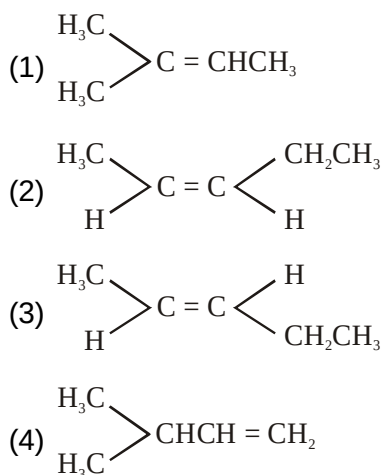
- (1)
(2)
(3) $\text{CH}_3\text{CH=CHCH}_3$
(4) $\text{CH}_3\text{CH}_2\text{CH=CH}_2$

62. $\text{A} \xrightarrow{\text{O}_3/\text{Zn}, \text{H}_2\text{O}} (\text{CH}_3)_2\text{CO} + \text{HCHO} + \begin{matrix} \text{CHO} \\ | \\ \text{CHO} \end{matrix}$

Thus, A is

- (1) $(\text{CH}_3)_2\text{CHCH=CHCH=CH}_2$
(2) $(\text{CH}_3)_2\text{C=CHCH=CH}_2$
(3) $\text{CH}_3\text{CH=CHCH=CHCH=CH}_2$
(4) None of the above

63. Which is the most stable alkene?



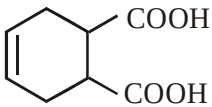
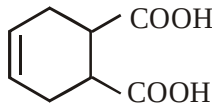
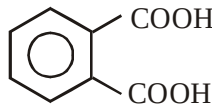
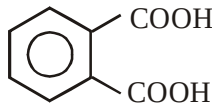
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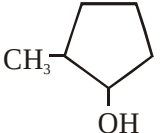
64. One mole of alkene X on ozonolysis gave one mole of acetaldehyde and one mole of acetone. The IUPAC name of X is

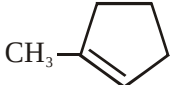
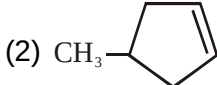
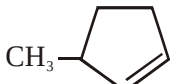
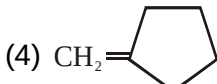
- (1) 2-methyl-2-butene
- (2) 2-methyl-1-butene
- (3) 2-butene
- (4) 1-butene

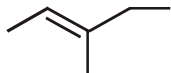
65. $\text{CH}_2=\text{CH}-\text{CH}=\text{CH}_2 + \begin{array}{c} \text{CHOOH} \\ \parallel \\ \text{CHOH} \end{array} \longrightarrow$

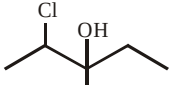
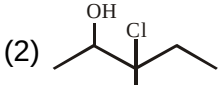
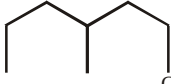
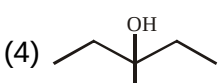
Product X by reaction R; X and R are

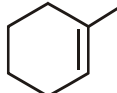
- (1)  Diels-Alder
- (2)  Friedle-Crafts
- (3)  Diels-Alder
- (4)  Friedle-Crafts

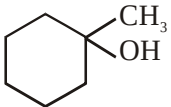
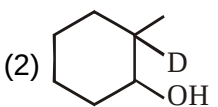
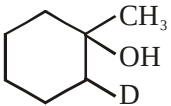
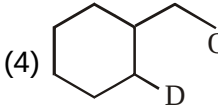
66.  $\xrightarrow[\text{heat}]{\text{H}_2\text{SO}_4}$ A (major), A is

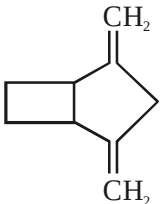
- (1) 
- (2) 
- (3) 
- (4) 

67.  $\xrightarrow[\text{H}_2\text{O}]{\text{Cl}_2}$ A, A is

- (1) 
- (2) 
- (3) 
- (4) 

68.  $\xrightarrow[\text{H}_2\text{O}_2/\text{OH}^-]{\text{BD}_3/\text{THF}}$ Product A, A is

- (1) 
- (2) 
- (3) 
- (4) 

69. Degree of unsaturation in  is

- (1) 2
- (2) 3
- (3) 4
- (4) 5

70. Which one is most easily dehydrohalogenated?

- (1) I
- (2) II
- (3) III
- (4) All with same ease

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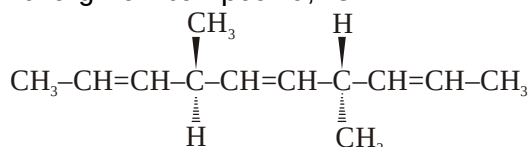
71. $\text{CH}_2=\text{CHCH}_2\text{CH}=\text{CH}_2 \xrightarrow{\text{NBS}}$ A, A is

- (1) $\text{CH}_2=\text{CHCH}(\text{Br})\text{CH}=\text{CH}_2$
- (2) $\text{CH}_2=\text{CHC}(\text{Br})=\text{CHCH}_2\text{Br}$
- (3) $\text{CH}_2=\text{CHCH}_2\text{CH}=\text{CHBr}$
- (4) $\text{CH}_2=\text{CHCH}_2\text{C}(\text{Br})=\text{CH}_2$

72. $\text{CH}_3\text{CH}(\text{C}_6\text{H}_5)\text{CH}=\text{CH}_2 \xrightarrow{\text{A}}$ $\text{CH}_3\text{CH}(\text{C}_6\text{H}_5)\text{CH}(\text{OH})\text{CH}_3$ A can be

- (1) $\text{H}_2\text{O}/\text{H}^+$
- (2) $\text{BH}_3 \cdot \text{THF}/\text{H}_2\text{O}_2-\text{OH}^-$
- (3) $\text{Hg}(\text{OCOCH}_3)_2 \cdot \text{THF}/\text{NaBH}_4 \cdot \text{NaOH}$
- (4) All of the above are true

73. The number of optically active products obtained from the complete ozonolysis of the given compound, is



- (1) 0
- (2) 1
- (3) 2
- (4) 4

74. Ethynic hydrogen is released most easily in

- (1)
- (2)
- (3)
- (4)

75. Maximum hyperconjugation is observed in

- (1)
- (2)
- (3) $(\text{CH}_3)_3\text{C}-\text{CH}=\text{CH}_2$
- (4)

76. In the following, electrophile of $\text{HO}-\text{NO}_2$ is

- (1) $\text{H}^\oplus$
- (2) $\text{NO}_3^\oplus$
- (3) $\text{NO}_2^\oplus$
- (4) $\text{OH}^\oplus$

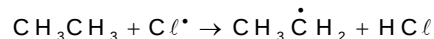
77. Nucleophilic addition will be maximum in

- (1) HCHO
- (2) $\text{C}_6\text{H}_5\text{CHO}$
- (3) CH_3CHO
- (4) CH_3COCH_3

78. In which proxide effect is not observed?

- (1) HF
- (2) HCl
- (3) HI
- (4) in all cases

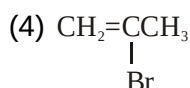
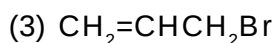
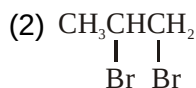
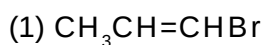
79. Following step is



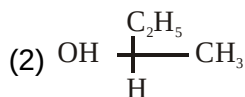
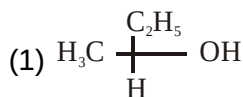
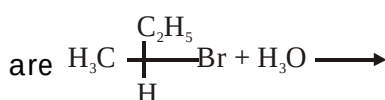
- (1) chain initiation
- (2) chain propagation
- (3) chain termination
- (4) chain disproportionation

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80. $\text{CH}_2=\text{CHCH}_3$ undergoes free radical substitution using NBS. Product formed is



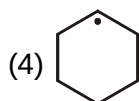
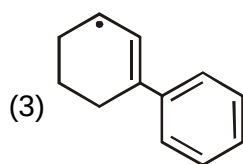
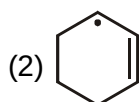
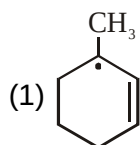
81. Product(s) of the following $\text{S}_\text{N}1$ reaction is/



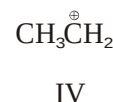
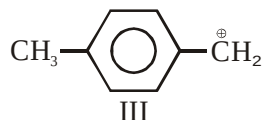
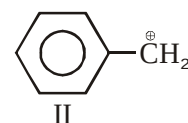
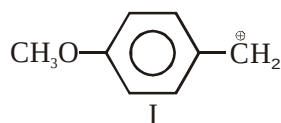
(3) Both (a) and (b)

(4) None of these

82. Most stable free radical is



83. Relative stabilities of the following carbocation will be in the order



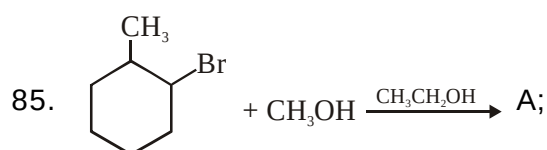
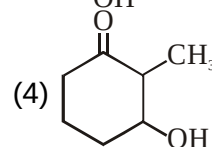
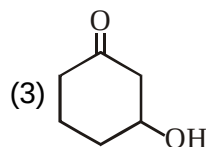
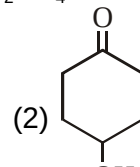
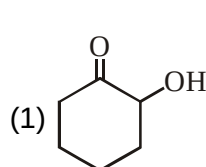
(1) $\text{I} < \text{II} < \text{III} < \text{IV}$

(2) $\text{IV} < \text{III} < \text{II} < \text{I}$

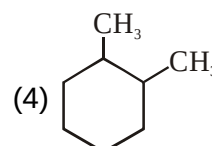
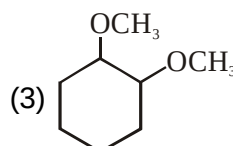
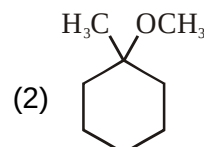
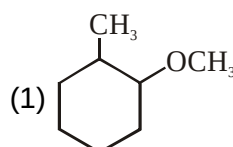
(3) $\text{IV} < \text{II} < \text{III} < \text{I}$

(4) $\text{II} < \text{IV} < \text{III} < \text{I}$

84. Which is dehydrated to a maximum extent using conc. H_2SO_4 ?

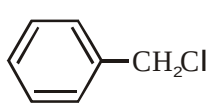


A (predominant A) is

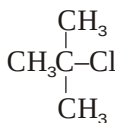


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86. There is formation of precipitate almost instantly when alcoholic AgNO_3 reacts with



I



II



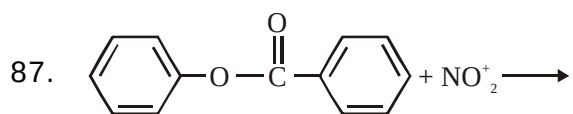
III

(1) I, II and III

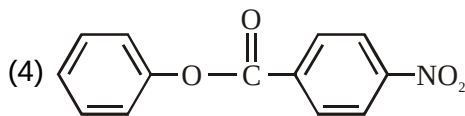
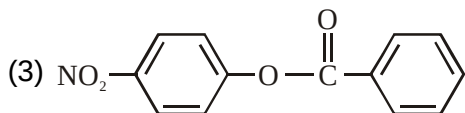
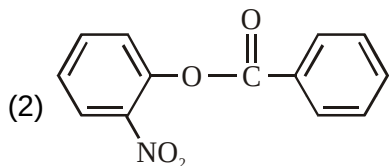
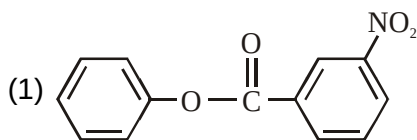
(2) I and II

(3) II and III

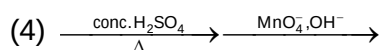
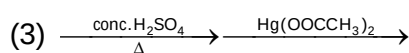
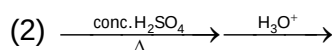
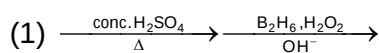
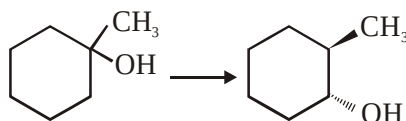
(4) I and III



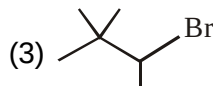
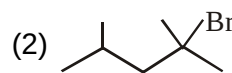
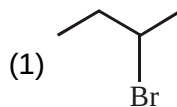
Product of this reaction by single S_E reaction



88. Provide a sequence of reactions to carry out the following change



89. Which of the following alkyl halides would be most likely to give a rearranged product under $\text{S}_\text{N}1$ conditions?



90. Which one of the following C-H bonds is the weakest for homolytic fission?

(1) $\text{CH}_3\text{-H}$

(2) $(\text{C}_6\text{H}_5)\text{CH}_2\text{-H}$

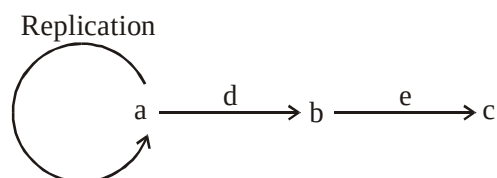
(3) $\text{C}_6\text{H}_5\text{-H}$

(4) $(\text{C}_6\text{H}_5)_3\text{C-H}$

SPACE FOR ROUGH WORK

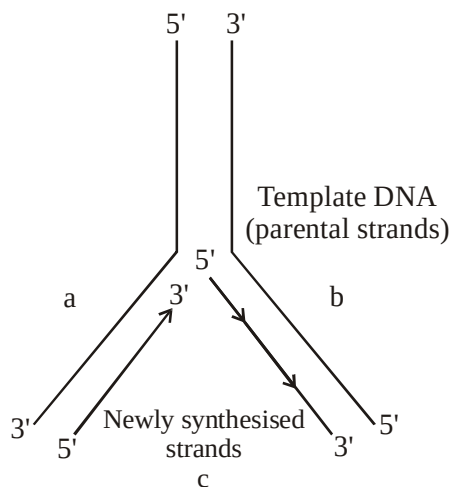
91. The enzyme that removes the RNA primers is called
 (1) DNA ligase
 (2) Primase
 (3) helicase
 (4) DNA polymerase I
135. Purines of DNA are represented by
 (1) Uracil and thymine
 (2) Guanine and adenine
 (3) Uracil and cytosine
 (4) Thymine and cytosine
93. In quantitative inheritance F_2 ratio obtained in a dihybrid cross is
 (1) 1 : 4 : 6 : 4 : 1 (2) 15 : 1
 (3) 12 : 3 : 1 (4) 9 : 7
94. The energy necessary for making a DNA molecule comes directly from the
 (1) sugar
 (2) ATP
 (3) release of phosphates
 (4) NADPH
95. When a certain character is inherited only through female parents, it probably represents
 (1) Multiple plastid inheritance
 (2) Cytoplasmic inheritance
 (3) Incomplete dominance
 (4) Mendelian nuclear inheritance
96. In eukaryotic cells, each chromosome has
 (1) one origin of replication
 (2) two origins of replication
 (3) many origins of replication
 (4) Only one origin of replication per nucleus
97. Nucleosome is :-
 (1) Intron interrupted DNA
 (2) Double helix DNA
 (3) Negatively charged DNA wrapped around positively charged histone octamer
 (4) Satellite DNAs

98. Genetic material is :
 (1) any nucleic acid
 (2) a nucleic acid that is capable of being used as a template in replication
 (3) a nucleic acid that is capable of being used as a template in transcription
 (4) a nucleic acid that is capable of being used as a template in replication and transcription
99. Diagram represents "central dogma" of molecular biology. Choose correct combination of labelling.



- (1) a-Protein, b-RNA, c-DNA, d-Translation, e-Transcription
 (2) a-DNA, b-RNA, c-Protein, d-Transcription, e-Translation
 (3) a-RNA, b-DNA, c-Protein, d-Transcription, e-Translation
 (4) a-Transcription, b-Translation, c-Protein, d-DNA, e-RNA
100. A DNA nucleotide chain has AGCTTCGA sequence. The nucleotide sequence of other chain would be :-
 (1) TCGAAGCT
 (2) GCTAAGCT
 (3) TAGCATAT
 (4) GATCCTAG
101. Information transfer from RNA to DNA is :-
 (1) Transcription
 (2) Translation
 (3) Replication
 (4) Reverse transcription

102. Recognise the figure and find out the correct matching.



- (1) a–continuous synthesis,
b–discontinuous synthesis,
c–replication fork
- (2) a–discontinuous synthesis,
b–continuous synthesis,
c–okazaki fragments
- (3) a–continuous synthesis,
b–discontinuous synthesis,
c–okazaki fragments
- (4) a–template strand,
b–coding strand,
c–replication fork

103. Process used by Meselson and Stahl for studying semi-conservative replication of DNA was :-

- (1) Chromatography
- (2) Density gradient centrifugation
- (3) Buoyant density centrifugation
- (4) Centrifugation

104. Which of the following features summarizes the molecular architecture of DNA?

- (1) The two strands run in opposite directions
- (2) The molecule twists in the same direction as the threads of most screws
- (3) The molecule is a double-stranded helix and DNA has a uniform diameter
- (4) All of the above

105. A DNA strand is directly involved in the synthesis of all the following except :

- (1) t-RNA molecule
- (2) m-RNA molecule
- (3) another DNA strand
- (4) protein synthesis

106. Semi-conservative mode of DNA replication in eukaryotes was proved by :-

- (1) Taylor
- (2) Meselson and Stahl
- (3) Watson
- (4) Crick

107. Which feature of the Watson-Crick model of DNA structure explains its ability to function in replication and gene expression?

- (1) Each strand contains all the information present in the double helix
- (2) Structural and functional similarities of DNA and RNA
- (3) The double helix is right-handed and not left-handed
- (4) DNA replication does not require enzyme catalysts

108. In polygenic inheritance :-

- (1) $\frac{n(n+1)}{2}$ formula can be used to find out total possible genotypes
- (2) $(2n+1)$ formula can be used to find out total possible phenotypes.
- (3) Number of intermediates increases with decrease in number of genes.
- (4) Mendel is considered as founder of polygenic inheritance.

109. A segment of DNA molecule contains 20,000 bases. What is the number of phosphate molecule in this DNA molecule ?

- (1) 20,000
- (2) 40,000
- (3) 10,000
- (4) 5,000

110. Select the set of correct statements:

- A–Coplanarity of stacks of nucleobases, confers additional stability to DNA double helix.
- B–Mendel succeeded as he has taken small sampling size for experiments.
- C–Mendelian factors are actually genes.
- D–ABO Blood group is an example of polygenic inheritance.

- (1) A, B
- (2) A, C
- (3) B, C
- (4) C, D

111. Observe the phenotypic effects of a gene with 2 alleles—B and b in pea plant.

Genotype	Grain Size	Grain Shape
BB	Large	Round
Bb	Intermediate	Round
bb	Small	Wrinkled

Which of the following extract is correct in concern to above facts?

- (1) Gene shows co-dominance
- (2) Gene shows incomplete dominance for grain shape.
- (3) Gene is example of epistatic gene.
- (4) Gene shows incomplete dominance for grain size.

112. Match the columns:

Column A		Column B	
A	Taylor and colleagues	(i)	Biochemical characterization of transforming principle
B	Meischer	(ii)	Nuclein
C	Crick	(iii)	Central dogma
D	Avery, Macleod and McCarthy	(iv)	Semi-conservative replication of DNA

- (1) A—(iv), B—(ii), C—(i), D—(iii)
- (2) A—(iv), B—(ii), C—(iii), D—(i)
- (3) A—(ii), B—(iv), C—(i), D—(iii)
- (4) A—(iv), B—(i), C—(iii), D—(ii)

113. Discontinuous fragments of DNA which are formed during replication of DNA are joined by:

- (1) Ligase
- (2) Lyase
- (3) DNA polymerase
- (4) RNA polymerase

114. A nucleoside differs from a nucleotide in not having

- (1) Phosphate
- (2) Sugar
- (3) Nitrogen base
- (4) Phosphate and sugar

115. The force that holds DNA together in a double helix is

- (1) The force of the twist
- (2) covalent bonds
- (3) ionic bonds or ionic interactions
- (4) hydrogen bonds

116. The ratio of childrens with blood groups O : AB : B : A born to a set of parents in which mother is with B blood group and father with A blood group, will be :-

- (1) 1 : 1 : 1 : 1
- (2) 2 : 0 : 2 : 0
- (3) 0 : 0 : 4 : 0
- (4) All of these

117. $\phi \times 174$ bacteriophage contains :-

- (1) ss DNA
- (2) ds DNA
- (3) ds RNA
- (4) ss RNA

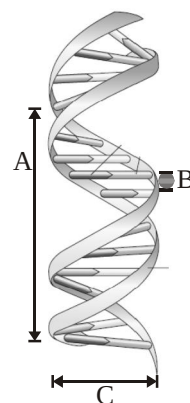
118. Read the following 4 statement (a-d) :-

- (a) Nucleosome contain 200 base pair of DNA
- (b) Nucleosome = Binding DNA + Histone Octamer + Linker DNA + H_1 histone
- (c) 6 Nucleosome units united to form Solenoid structure
- (d) Histone octamer contain $(H_2A \ H_2B \ H_3H_1)_2$

How many above statement are correct ?

- (1) 4
- (2) 3
- (3) 2
- (4) 1

119. Given figure represents the DNA double helix model as proposed by Watson and Crick (1953). Select the option that shows correct measurements of A, B and C.



- (1) A-3.4 nm, B – 0.34 nm, C – 2 nm
- (2) A-34 nm, B-3.4 nm, C – 20 nm
- (3) A – 3.4 Å, B – 0.34 Å, C – 20 Å
- (4) A – 34 Å, B – 3.4 Å, C – 2 Å

120. Which of the following is Nucleoside ?

- (1) Adenylic Acid
- (2) Adenosine, Uridine
- (3) Adenine, Uracil
- (4) Deoxy, Uridine

121. 5 Methyl uracil is present in which Nucleic Acid
 (1) RNA and DNA
 (2) Only DNA
 (3) Only RNA
 (4) Only double helical RNA
122. Choose correct statement from Griffith experiment
 (1) S strain → inject in to mice → Mice died
 (2) R strain → inject in to mice → mice live
 (3) S strain (Heat killed) → Inject in to mice → Mice die
 (4) Both 1 and 2
123. Identify the correctly matched pair.
 (1) $\phi \times 174$ – 5836 Nucleotide
 (2) Bacteriophage – 48502 BP
 (3) *E. Coli* – 4.6×10^{12} BP
 (4) Human genome – 3.3×10^6 BP
124. In the B-DNA, number of nitrogenous base pairs in 10 turns:
 (1) 10 (2) 20 (3) 50 (4) 100
125. In the bacteriophage $\phi \times 174$ number of nucleotide present are 5386. Calculate the number of phospho-ester bond and phosphate group sugar in it.
 (1) 5385, 5386 (2) 5386, 5386
 (3) 1, 5385 (4) 1, 5386
126. Which of the following findings are not associated with work of Chargaff on DNA?
 (1) $\frac{T}{A} = 1$, follow Chargaff rule
 (2) $\frac{A+G}{T+C} = 1$, this value is constant for all species
 (3) Base pair ratio can be used for identification of species
 (4) Purines and pyrimidines occur in equal amount in one strand
127. Kornberg enzyme is :-
 (1) DNAP - I (2) DNAP - II
 (3) DNAP - III (4) All of these
128. Read the following statements carefully w.r.t. DNA replication:
 (i) The main enzyme is DNA dependent DNA polymerase.
 (ii) Deoxyribonucleoside monophosphates provides energy for polymerisation.
- (ii) Polymerisation occur only in one direction that is $5' \rightarrow 3'$.
 (iv) Prokaryotes have single 'ori' site.
 (v) Viruses have single 'ori' site.
 (vi) In Eukaryotes, the replication of DNA takes place at S phase of cell cycle.
 (vii) In *E. coli*, the replication of DNA takes place in any stage of interphase.
 How many of the above given statements are correct?
 (1) three (2) four
 (3) five (4) six
129. Which of the following character is not applicable to tRNA ?
 (1) It is the smallest of the RNAs
 (2) It acts as an adaptor for amino acid
 (3) It has a clover leaf like structure
 (4) It is the largest of the RNAs
130. Which of the following rRNAs acts as structural RNA as well as ribozyme in bacteria?
 (1) 5 S rRNA (2) 5.8 S rRNA
 (3) 18 S rRNA (4) 23 S rRNA
131. Which one of the following generates new genetic combinations leading to variation?
 (1) Segregation of alleles
 (2) Independent assortment
 (3) Fertilisation
 (4) All of above
132. If a double helix DNA is kept at 90°C , then
 (1) It changes into RNA
 (2) It divides into one million pieces
 (3) No effect
 (4) It uncoils into two strands
133. How many types of ribonucleic acids are known
 (1) One type (2) Two types
 (3) Three types (4) Four types
134. Ribozyme is
 (1) messenger RNA (2) transfer RNA
 (3) Catalytic RNA (4) Structural RNA
135. Mendel did not recognise the linkage phenomenon in his experiment because :
 (1) He did not have powerful microscope
 (2) He studied only pure lines
 (3) Genes of characters he studied present on different chromosomes
 (4) More than one option is correct

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| <p>136. CDRI stands for</p> <ol style="list-style-type: none"> (1) Central Dairy Research Institute (2) Central Drug Related Institute (3) Central Drug Research Institute (4) Central Developmental Research Institute <p>137. "Saheli", an oral contraceptive for females, was developed by</p> <ol style="list-style-type: none"> (1) AIIMS, Delhi (2) IICB, Kolkata (3) SGPGI, Lucknow (4) CDRI, Lucknow <p>138. A contraceptive method in which the male partner withdraws his penis from the vagina just before ejaculation as to avoid insemination, is called</p> <ol style="list-style-type: none"> (1) Periodic abstinence (2) Coitus interruptus (3) Withdrawal method (4) Both (2) and (3) <p>139. 'Nirodh' is a popular brand of</p> <ol style="list-style-type: none"> (1) Contraceptive pill for the females (2) Emergency IUDs for females (3) Condoms for females (4) Condoms for males <p>140. Spermicidal creams, jellies and foams are usually used along with the barrier methods to</p> <ol style="list-style-type: none"> (1) Increase their conception efficiency (2) Stimulate their conceptive efficiency (3) Increase their contraceptive efficiency (4) Provide motility to sperms <p>141. Diaphragms, cervical caps and vaults prevent conception by</p> <ol style="list-style-type: none"> (1) Increasing phagocytosis of sperms within uterus (2) Suppressing sperm motility (3) Inhibiting ovulation and implantation (4) Blocking the entry of sperms through the cervix | <p>142. Cu ions released from the copper releasing IUDs to</p> <ol style="list-style-type: none"> a. Suppress sperm motility b. Make uterus unsuitable for implantation c. Suppress fertilising capacity of sperms d. Make cervix hostile to sperms <ol style="list-style-type: none"> (1) a and b (2) b and d (3) a and c (4) a, c and d <p>143. In tubectomy, a small part is cut or tied up</p> <ol style="list-style-type: none"> (1) Vagina (2) Uterus (3) Abdomen (4) Fallopian tube <p>144. The contraceptive device which is highly effective but reversibility is very poor, is</p> <ol style="list-style-type: none"> (1) Barrier method (2) Surgical method (3) IUDs (4) Traditional method <p>145. A contraceptive method in which the couples avoid or abstain from coitus from day 10 to 17 of the menstrual cycle when ovulation could be expected, is called</p> <ol style="list-style-type: none"> (1) Lactational amenorrhea (2) Periodic abstinence (3) Coitus interruptus (4) Multiload <p>146. Intentional or voluntary termination of pregnancy before full term is called</p> <ol style="list-style-type: none"> (1) MTP (2) STD (3) Induced abortion (4) Both (1) and (3) <p>147. Hepatitis-B and HIV can be transmitted from one person to other by -</p> <ol style="list-style-type: none"> (1) Sharing of injection needles, surgical instruments, etc., with infected person (2) Transfusion of blood (3) From infected mother to the foetus (4) All of the above <p>148. Inability to conceive or produce children even after 1 years of unprotected sexual intercourse is called</p> <ol style="list-style-type: none"> (1) Sexuality (2) ART (3) Fertility (4) Infertility |
|---|---|

149. The infertile couples could be assisted to have children through certain special techniques commonly called as
 (1) RCH (2) ART
 (3) MTP (4) RTI
150. Female who cannot produce ovum, but can provide suitable environment for fertilisation and further development, could be assisted by
 (1) ZIFT (2) GIFT
 (3) ICSI (4) IUI
151. IVF followed by ET is called
 (1) Family planning programme
 (2) RCH programme
 (3) RTI
 (4) Test-tube baby programme
152. Stage transferred to uterus after induced fertilization of ova in the laboratory is
 (1) Embryo at two-blastomere stage
 (2) Morula
 (3) Zygote
 (4) Embryo at more than 16 blastomere stage.
153. Donor semen is introduced into uterus through
 (1) Intrauterine transfer (IUT)
 (2) Intrauterine insemination (IUI)
 (3) Gamete intrafallopian transfer (GIFT)
 (4) Intracytoplasmic sperm injection (ICSI)
154. In which ART technique, the semen is artificially introduced into the female?
 (1) ET (2) IUI
 (3) IUT (4) GIFT
155. ZIFT is transfer of
 (1) Zygote into fallopian tube
 (2) Embryo into uterus
 (3) Mixture of sperm and ova into fallopian tube
 (4) Mixture of sperms and ova into uterus
156. Artificial insemination means
 (1) Artificial introduction of sperms of a healthy donor into the vagina
 (2) Introduction of sperms of healthy donor directly into the ovary
 (3) Transfer of sperms of a healthy donor to a test tube containing ova
 (4) Transfer of sperms of husband to a test tube containing ova
157. Induced abortion is
 (1) IUD (2) PID
 (3) MTP (4) STD
158. Which contraceptive device makes uterus unsuitable for implantation
 (1) Progestasert (2) CuT
 (3) Lippe's loop (4) Multiload 375
159. Ectopic pregnancies are referred to as
 (1) Implantation of embryo at site other than uterus
 (2) Implantation of defective embryo in the uterus
 (3) Pregnancies terminated due to hormonal imbalance
 (4) Pregnancies with genetic abnormality
160. The hormone releasing intrauterine devices such as Progestasert specifically
 (1) Suppress sperm motility
 (2) Increase phagocytosis of sperms
 (3) Make uterus unsuitable for implantation
 (4) Suppress fertilizing capacity of sperms
161. In India, the family planning program was launched in:
 (1) 1977 (2) 1951
 (3) 1967 (4) 1981
162. Prenatal genetic defects in foetus can be detected by:
 (1) CT-scan
 (2) X-ray radiography
 (3) MRI
 (4) Amniocentesis

163. Which of the following is the cause of population explosion in India?

- (1) Rapid decline in death rate
- (2) Decline in mortality maternal rate (MMR)
- (3) Increase number of people in reproductive age
- (4) All of these

164. Emergency contraceptives are effective if used within:

- (1) 72 hours of fertilization
- (2) 72 hours of coitus
- (3) 72 hours of implantation
- (4) 72 hours of ovulation

165. Match the contraceptive methods given under Column I with their examples given under Column II. Select the correct choice from those given below.

Column-I		Column-II	
A.	Barrier device	p.	Depo-Provera
B.	Sterilization	q.	Condom, Diaphragm
C.	IUDs	r.	Multiload-375
D.	Injectable contraceptive	s.	Vasectomy

- (1) A-q, B-p, C-r, D-s
- (2) A-f, B-s, C-p, D-q
- (3) A-q, B-s, C-r, D-p
- (4) A-s, B-q, C-p, D-r

166. Zygote or early embryo upto eight blastomeres is transferred into fallopian tube by a procedure called:

- (1) Intra Uterine Transfer (IUT)
- (2) Zygote Intra Fallopian Transfer (ZIFT)
- (3) Gamete Intra Fallopian Transfer (GIFT)
- (4) Intra Uterine Insemination (IUI)

167. Which of the following statement is correct regarding MTP?

- (1) 60-65 million pregnancies are terminated world wide every year.
- (2) One fifth of the total conceived pregnancies are terminated.
- (3) MTP can be used for contraception.
- (4) MTPs are safe upto 20 weeks.

168. Which of the following is incorrect regarding contraceptive devices?

- (1) Progestasert and LNG-20 are hormone releasing IUDs.
- (2) Spermicidal creams and jellies increase the efficiency of barrier devices.
- (3) Copper-T is placed in vagina and can be self inserted.
- (4) Condoms are useful in protecting against STDs.

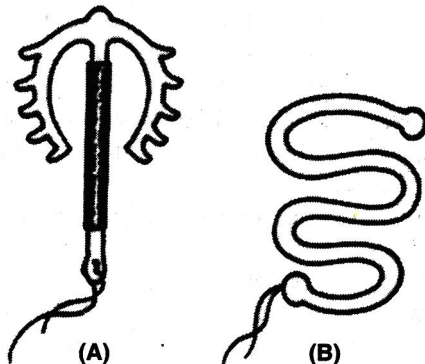
169. Which one of the following statements is wrong?

- (1) The national 'family planning program' was launched in India in 1951.
- (2) MTPs are considered relatively safe upto second trimester.
- (3) Contraceptive methods are commonly used to prevent unwanted pregnancies.
- (4) In India, Nirodh is popularly marketed condom made of latex.

170. Which one of the following statements is incorrect?

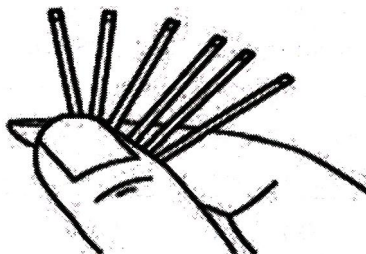
- (1) Medical termination of pregnancy is legalised in our country.
- (2) Coitus interruptus is one of the oldest forms of contraceptive methods known to man.
- (3) *In vitro* fertilization (IVF) is a simple, cost-effective procedure for infertile couples.
- (4) Saheli is the world's first non-steroidal oral contraceptive pills.

171. Two Intra Uterine Devices (IUD) are given below. Correctly identify them by selecting the option:



- (1) A - LNG -20, non-medicated IUD
B - Lippes loop, hormonal IUD
(2) A - Multiload 375, Cu-releasing IUD
B - Lippes loop, non-medicated IUD
(3) A - Lippes loop, hormonal IUD
B - Copper T, non-medicated IUD
(4) A - Lippes loop, hormonal IUD
B - Copper 7, non-medicated IUD

172. Identify the image of the device given below used as capsules surgically implanted under skin slowly release hormone that blocks ovulation?



- (1) Levonorgestrol (2) Progestasert
(3) Norplant (4) Depo-Provera

173. Diaphragms are contraceptive devices used by the females. Choose the correct options from the statements given below:

- i. They are introduced into the uterus
ii. They are placed to cover the cervical region
iii. They act as physical barriers for sperm entry
iv. They act as spermicidal agents

- (1) i and ii (2) i and iii
(3) ii and iii (4) iii and iv

174. Progesterone present in contraceptive pill is meant for

- (1) Checking ovulation
(2) Preventing fertilisation
(3) Preventing implantation of zygote
(4) Preventing cleavage

175. Tubectomy, a method of population control, is performed in

- (1) Both males and females
(2) Males only
(3) Females only
(4) Only pregnant females

176. Which of the following is a mechanical barrier used in birth control?

- (1) Copper T (2) Diaphragm
(3) Loop (4) LNG 20

177. Which one is the legal method of birth control?

- (1) Termination of unwanted pregnancy
(2) Sterilisation techniques
(3) The rhythm method
(4) Use of physical barriers

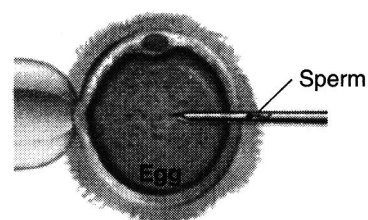
178. Purpose of tubectomy is to prevent

- (1) Coitus
(2) Egg formation
(3) Fertilization
(4) Sperm formation

179. Oral contraceptives are used by females to check

- (1) Implantation
(2) Fertilization
(3) Ovulation
(4) Entry of sperms in vagina

180. What is the figure given below showing one of the assisted reproductive technologies?



- (1) ZIFT (2) IUT
(3) GIFT (4) ICSI